Prediction Markets

Information. Prices. Outcomes.

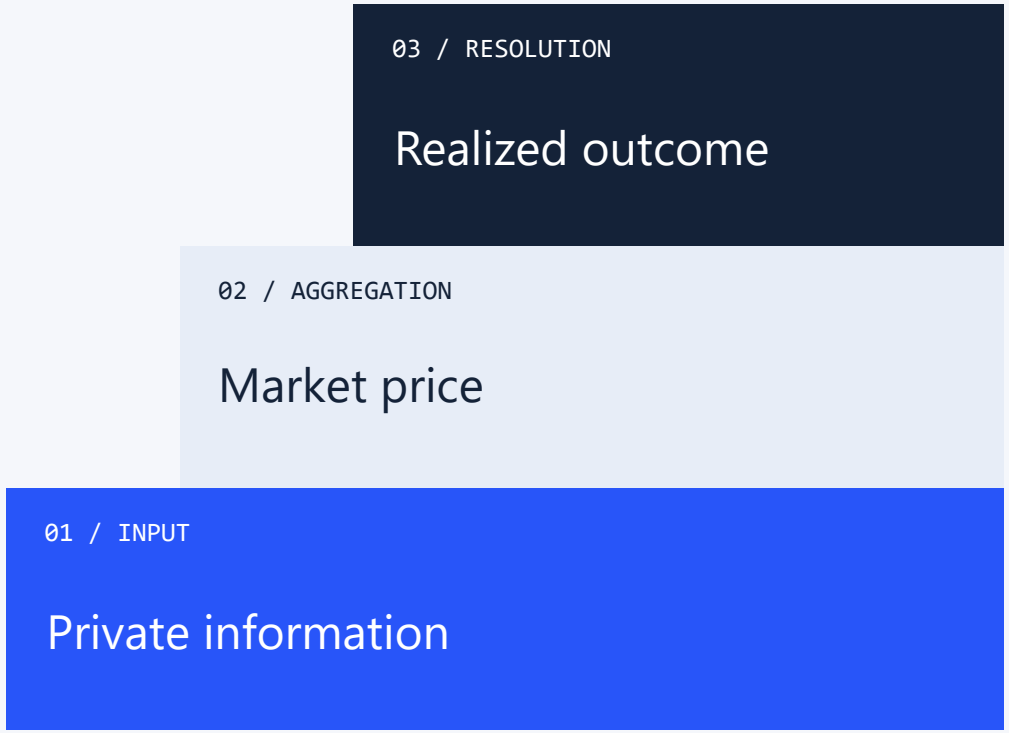
Presenter Name

Institution Name



Markets connect information to outcomes

- Define a verifiable event.
- Trade a contingent claim.
- Resolve against a stated rule.



Conceptual structure; the stepped geometry does not represent measured data.

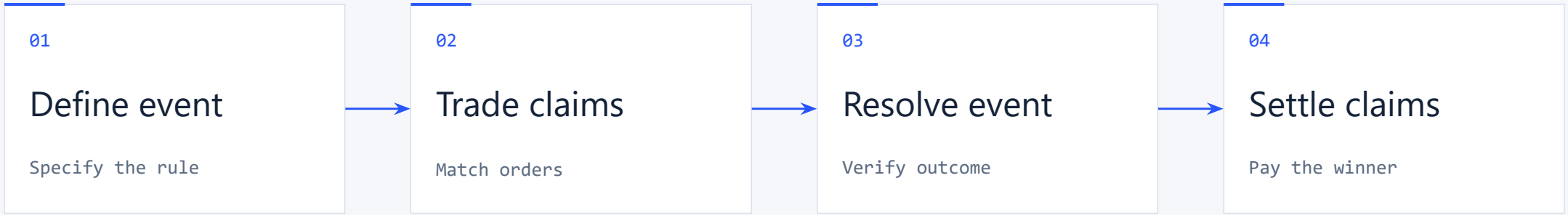
One dollar backs a complete outcome pair

1 YES + 1 NO ⇄ $1

- Split collateral into complementary claims. 01 / SPLIT
- Merge a complete pair to recover collateral. 02 / MERGE
- The winning claim pays one dollar. 03 / SETTLE

Illustrative fully collateralized binary contract.

A stated rule connects trading to settlement



- Specify the resolution rule before trading.
- Match orders while the market is open.
- Pay the winning claim after resolution.

Illustrative lifecycle for a fully collateralized binary market; platform procedures vary.

The best quotes define the trading spread

- The ask is an offer to sell.
- The bid is an offer to buy.
- Their difference is the spread.

LIMIT ORDER BOOK / SNAPSHOT

Side	Price	Shares
Ask	0.62	1,200
Ask	0.61	3,500
Ask	0.60	5,000
Bid	0.59	4,200
Bid	0.58	2,800

Spread 0.60 − 0.59 = 0.01

Synthetic order book for demonstrating the layout; not a live market snapshot.

New information can move the market price

- Signals change expected payouts.
- Orders transmit those signals.
- Prices adjust through trading.



Schematic price path with invented values; replace with a documented series.

Trading activity does not establish accuracy

- Volume measures participation.
- Spreads measure trading costs.
- Calibration requires outcomes.

01 / ACTIVITY

Volume

02 / LIQUIDITY

Bid–ask spread

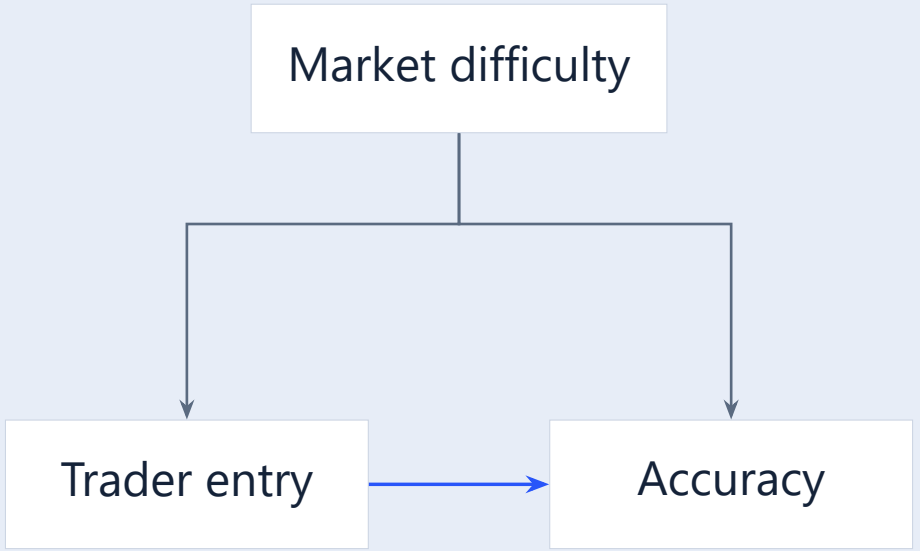
03 / ACCURACY

Forecast vs. outcome

Distinct measures answer distinct questions; no empirical ranking is implied.

Selection complicates causal interpretation

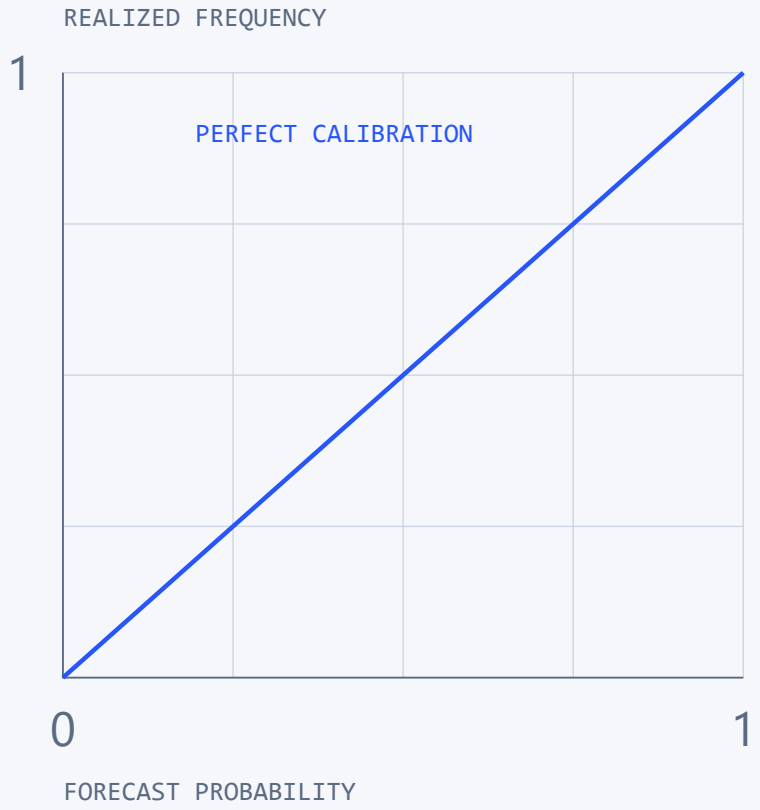
- Measure skill before the event.
- Separate entry from performance.
- Seek credible external variation.



Illustrative confounding structure; arrows express a research hypothesis.

Calibration compares forecasts with outcomes

- Group comparable forecasts.
- Compute realized frequencies.
- Report sampling uncertainty.



Reference diagonal only; no observations or estimated calibration curve are supplied.

Each question needs a corresponding test

01	Information costs	Credible cost shock
02	Trader composition	Variation beyond selection
03	Forecast accuracy	Realized outcomes

Template framework; replace these candidate questions with a specific research design.

Separate the price, the mechanism, and the evidence

- A price summarizes a tradable claim.
- Trading incentives shape that price.
- Accuracy remains an empirical question.

01 / PRICE

Contingent value

02 / MECHANISM

Information and incentives

03 / EVIDENCE

Outcomes and identification

The example is a research scaffold, not a completed empirical study.